CORE PURE / FURTHER CALCULUS

Integration with Inverse Trig

Mark scheme

17 questions · 138 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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	Total	**138**	

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Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$x(4-x) = 4 - (x-2)^2$	**M1**	Writes $x(4-x)$ in the form $p-(x-q)^2$. Condone slips in p or q.
	$\int \frac{1}{\sqrt{4-(x-2)^2}}\,\mathrm{d}x$ $= \arcsin\left(\frac{x-2}{2}\right)$	**M1 A1**	**M1** Integrates their $\frac{1}{\sqrt{p-(x-q)^2}}$ to the form $k\arcsin\left(\frac{x-q}{r}\right)$. Condone a wrong k or r. **A1** oe, such as $\arcsin\left(\frac{x}{2}-1\right)$ or $-\arccos\left(\frac{x-2}{2}\right)$.
	$\lim_{t\to 4^-} \arcsin\left(\frac{t-2}{2}\right) - \lim_{s\to 0^+} \arcsin\left(\frac{s-2}{2}\right)$ $= \arcsin(1) - \arcsin(-1)$	**dM1**	Evaluates their arcsin form as limits at 0 and 4 and subtracts. Not for substituting 0 and 4 with no limit seen.
	$= \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$	**A1**	
	Total for Question 1	**5**	

Question	**Scheme**	**Marks**	**Notes**
2	$(x^2-2x+5=)\ (x-1)^2+4$	**M1**	Completes the square to the form $(x\pm 1)^2+q$. Condone a wrong constant q.
	$\int_0^3 \frac{1}{(x-1)^2+2^2}\,\mathrm{d}x$ $=\frac{1}{2}\left[\arctan\left(\frac{x-1}{2}\right)\right]_0^3$	**M1 A1**	**M1** Integrates their $\frac{1}{(x\pm1)^2+q}$, $q>0$, to the form $k\arctan\left(\frac{x\pm1}{\sqrt{q}}\right)$, or to ku after a tan substitution. **A1** oe, such as $\frac{1}{2}u$ after $x-1=2\tan u$.
	$\frac{1}{2}\left(\arctan 1-\arctan\left(-\frac{1}{2}\right)\right)$	**dM1**	Substitutes 3 and 0 into their arctan form and subtracts, or uses the changed limits after a substitution.
	$=\frac{1}{2}\left(\frac{\pi}{4}+\arctan\frac{1}{2}\right)$	**A1**	Any correct exact form; implied by correctly combining $\arctan 1-\arctan\left(-\frac{1}{2}\right)$ in the next step.
	$\tan\left(\frac{\pi}{4}+\arctan\frac{1}{2}\right)$ $=\frac{1+\frac{1}{2}}{1-1\times\frac{1}{2}}=\dots$	**M1**	Applies $\tan(A\pm B)$, or the arctan addition formula, to their two angles with their tangents substituted, or shows $\arctan 3-\frac{\pi}{4}=\arctan\frac{1}{2}$. Condone sign slips.
	$=3$, with $\frac{\pi}{4}+\arctan\frac{1}{2}$ in $\left(0,\frac{\pi}{2}\right)$ $\Rightarrow \int_0^3 \frac{1}{x^2-2x+5}\,\mathrm{d}x=\frac{1}{2}\arctan 3$	**A1 AG**	Correct substituted formula leading without error to tangent 3, the angle placed in $\left(0,\frac{\pi}{2}\right)$, directly or as in $(0,\pi)$ with a positive tangent, and the printed result stated.
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3	$10+8x-2x^2=2\left(9-(x-2)^2\right)$	**M1**	Completes the square to the form $p\left(q-(x-r)^2\right)$ oe. Condone slips in the constants.
	$\frac{1}{\sqrt{2}}\int_1^4 \frac{1}{\sqrt{9-(x-2)^2}}\,\mathrm{d}x$ $=\frac{1}{\sqrt{2}}\left[\arcsin\left(\frac{x-2}{3}\right)\right]_1^4$	**M1 A1**	**M1** Takes out $\sqrt{p}$ and integrates to the form $k\arcsin\left(\frac{x-r}{s}\right)$, by the standard result or a substitution. Condone slips in k and s. **A1** oe, such as $\frac{1}{\sqrt{2}}\arcsin\frac{u}{3}$ with $u=x-2$.
	$=\frac{1}{\sqrt{2}}\left(\arcsin\left(\frac{2}{3}\right)-\arcsin\left(-\frac{1}{3}\right)\right)$	**dM1**	Substitutes both limits, or their transformed limits, into their arcsin form and subtracts.
	$=\frac{1}{\sqrt{2}}\left(\arcsin\left(\frac{2}{3}\right)+\arcsin\left(\frac{1}{3}\right)\right)$	**A1 AG**	Correct substituted limits leading without error to the printed result, with the sign of $\arcsin\left(-\frac{1}{3}\right)$ handled correctly.
	Total for Question 3	**5**	

Question	**Scheme**	**Marks**	**Notes**
4(a)	The upper limit is ∞: the interval of integration is unbounded.	**E1**	States the upper limit is infinite or the interval is unbounded. E0 if it also says the integrand is undefined in the interval.
		[1]	
4(b)	$x^2 - 6x + 18 = (x-3)^2 + 9$	**M1**	Writes the denominator as $(x \pm 3)^2 + c$ with $c > 0$.
	$\int \frac{1}{(x-3)^2+9}\,\mathrm{d}x = \frac{1}{3}\arctan\left(\frac{x-3}{3}\right)$	**M1 A1**	**M1** Integrates their completed square to the form $p\arctan(q(x-3))$, or $p\arctan(qu)$ after substituting $u = x - 3$.
	$\lim_{t\to\infty}\left[\frac{1}{3}\arctan\left(\frac{x-3}{3}\right)\right]_0^t$ $= \frac{1}{3}\left(\frac{\pi}{2} - \left(-\frac{\pi}{4}\right)\right)$	**dM1**	Applies limits 0 and t to their arctan form, using $\arctan \to \frac{\pi}{2}$ as $t \to \infty$. Condone ∞ substituted directly.
	$= \frac{\pi}{4}$, so $k = \frac{1}{4}$	**A1**	cso. Accept $\frac{\pi}{4}$ with k not stated; a decimal value scores A0.
		[5]	
Total for Question 4		**6**	

Question	**Scheme**	**Marks**	**Notes**
5	$x^2 - 4x + 8 = (x-2)^2 + 4$	**M1**	Writes the denominator as $(x \pm 2)^2 + k$, $k > 0$. Condone a slip in k.
	$\int_0^4 \frac{1}{(x-2)^2+2^2}\,\mathrm{d}x$ $= \left[\frac{1}{2}\arctan\left(\frac{x-2}{2}\right)\right]_0^4$	**M1 A1**	**M1** Integrates to $p\arctan\left(\frac{x-2}{q}\right)$, or $p\arctan\frac{u}{q}$ with $u = x-2$. Condone wrong p or q.
	$\frac{1}{2}\arctan(1) - \frac{1}{2}\arctan(-1)$	**dM1**	Substitutes both limits, 4 and 0 (or ± 2 for u), into their arctan form and subtracts.
	$= \frac{\pi}{4}$	**A1**	cso.
	Total for Question 5	**5**	

Question	Scheme	Marks	Notes
6	$\int_{-1}^{4} \frac{9}{(x-1)^2+4}\,\mathrm{d}x$ $= \left[k \arctan\left(\frac{x-1}{2}\right)\right]_{-1}^{4}$	**M1**	Integrates to the form $k \arctan(px+q)$, by recognition or a substitution such as $x-1 = 2\tan\theta$. Condone slips in k.
	$= \left[\frac{9}{2} \arctan\left(\frac{x-1}{2}\right)\right]_{-1}^{4}$	**A1**	oe, including the correct integral in θ.
	$\frac{9}{2}\left(\arctan\frac{3}{2} - \arctan(-1)\right)$	**dM1**	Substitutes 4 and -1, or the corresponding limits of their substitution, and subtracts the right way round.
	$= \frac{9}{2}\left(\arctan\frac{3}{2} + \frac{\pi}{4}\right)\ \ [= 7.957\ldots]$	**A1**	Exact, or awrt 7.96.
	$V = 36 \times \text{“}7.957\text{”} \times 240$	**M1**	Multiplies their area in graph units by 6^2 and by 240, or works in centimetres throughout. Not for scaling by 6 alone.
	$= 6.87 \times 10^4\,\mathrm{cm}^3$	**A1**	awrt 6.87×10^4 (68 700), or awrt $0.0687\,\mathrm{m}^3$.
	Total for Question 6	**6**	

Question	Scheme	Marks	Notes
7	$u = \ln x \Rightarrow \mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$	**M1**	Uses $u = \ln x$ with $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$ oe to replace $\frac{1}{x}\,\mathrm{d}x$. Implied by integrating directly to an arcsin of a multiple of $\ln x$.
	$\int \frac{1}{\sqrt{4-u^2}}\,\mathrm{d}u$	**A1**	Limits need not be seen. Implied by $\arcsin\left(\frac{\ln x}{2}\right)$.
	$\arcsin\left(\frac{u}{2}\right)$	**M1 A1**	**M1** Integrates to $k\arcsin\left(\frac{u}{a}\right)$ or $k\arcsin(bu)$, any nonzero constants. **A1** Correct, or $\arcsin\left(\frac{\ln x}{2}\right)$.
	$x = 1, e \Rightarrow u = 0, 1$ $\arcsin\left(\frac{1}{2}\right) - \arcsin 0$	**dM1**	Applies u-limits 0 and 1, or x-limits 1 and e after substituting back, and subtracts.
	$= \frac{\pi}{6}$	**A1**	cao. Not 0.524 or 30°.
	Total for Question 7	**6**	

Question	**Scheme**	**Marks**	**Notes**
8(i)	$9x^2+12x+20=9\left(x+\frac{2}{3}\right)^2+\ldots$	**M1**	Writes the denominator as $9\left(x\pm\frac{2}{3}\right)^2+k$ or $(3x\pm 2)^2+k$, $k\neq 0$, oe.
	$=9\left(x+\frac{2}{3}\right)^2+16$	**A1**	oe, such as $(3x+2)^2+16$.
	$\int\ldots\,\mathrm{d}x=\frac{1}{9}\cdot\frac{1}{\text{“}\frac{4}{3}\text{”}}\arctan\left(\frac{x+\frac{2}{3}}{\text{“}\frac{4}{3}\text{”}}\right)$	**M1**	Integrates their completed square to $p\arctan(\text{linear in }x)$. Condone an incorrect p.
	$\frac{1}{12}\arctan\left(\frac{3x+2}{4}\right)+c$	**A1**	oe. Needs $+c$.
		[4]	
8(ii)	$45+12x-9x^2=\ldots-9\left(x-\frac{2}{3}\right)^2$	**M1**	Writes the quadratic as $k-9\left(x\pm\frac{2}{3}\right)^2$ or $k-(3x\pm 2)^2$, $k>0$, oe.
	$=49-(3x-2)^2$	**A1**	oe, such as $9\left[\frac{49}{9}-\left(x-\frac{2}{3}\right)^2\right]$.
	$\int\ldots\,\mathrm{d}x=\frac{1}{3}\arcsin\left(\frac{x-\frac{2}{3}}{\text{“}\frac{7}{3}\text{”}}\right)$	**M1**	Integrates their completed square to $p\arcsin(\text{linear in }x)$. Condone an incorrect p.
	$\frac{1}{3}\arcsin\left(\frac{3x-2}{7}\right)+c$	**A1**	oe. Needs $+c$.
		[4]	
Total for Question 8		**8**	

Question	Scheme	Marks	Notes
9(a)	$6x+1=\lambda(8x-8)+\mu \Rightarrow \lambda=\dots,\ \mu=\dots$	**M1**	Writes $6x+1$ as a multiple of $8x-8$ plus a constant, or splits the integral equivalently. Condone slips in the constants.
	$\frac{3}{4}\ln(4x^2-8x+13)$	**A1**	Or $A=\frac{3}{4}$ stated.
	$4x^2-8x+13=(2x-2)^2+9$ “7” $\int \frac{1}{(2x-2)^2+9}\,\mathrm{d}x = k\arctan\left(\frac{2x-2}{3}\right)$	**M1**	Completes the square and integrates their constant term to $k\arctan$ of a linear expression in x. Condone slips in k.
	$\frac{7}{6}\arctan\left(\frac{2x-2}{3}\right)$	**A1**	Or $B=\frac{7}{6}$ stated. Condone a missing $+c$.
		[4]	
9(b)	$\frac{1}{\frac{5}{2}-1}\int_1^{5/2} f(x)\,\mathrm{d}x$	**M1**	Divides the integral of $f(x)$ from 1 to $\frac{5}{2}$ by $\frac{3}{2}$, or multiplies it by $\frac{2}{3}$.
	$\frac{2}{3}\left(\text{“}\frac{3}{4}\ln 18+\frac{7}{6}\cdot\frac{\pi}{4}\text{”}-\text{“}\frac{3}{4}\ln 9\text{”}\right)$	**dM1**	Substitutes both limits into their (a) of the form $A\ln(4x^2-8x+13)+B\arctan\left(\frac{2x-2}{3}\right)$ and subtracts.
	$\frac{2}{3}\left(\frac{3}{4}\ln 2+\frac{7\pi}{24}\right)$ $=\frac{1}{2}\ln 2+\frac{7\pi}{36}$	**A1 AG**	Correct substituted expression leading without error to the printed result; combined steps accepted. A decimal check earns A0.
		[3]	
9(c)	$3-\ln 2-\frac{7\pi}{18}$	**B1**	oe, or awrt 1.09.
		[1]	
	Total for Question 9	**8**	

Question	Scheme	Marks	Notes
10	$4x - 1 = 2(2x - 4) + 7$	**M1**	Writes the numerator as a multiple of $2x - 4$ plus a constant, or splits the integrand equivalently. Condone arithmetic slips.
	$\int \frac{2(2x-4)}{x^2-4x+8}\,\mathrm{d}x$ $= 2\ln(x^2 - 4x + 8)$	**A1**	Correct term, or $B = 2$ stated. Condone a modulus.
	$x^2 - 4x + 8 = (x - 2)^2 + 4$ $\int \frac{\text{“}7\text{”}}{(x-2)^2+4}\,\mathrm{d}x$ $= \text{“}7\text{”} \times \frac{1}{2} \arctan\left(\frac{x-2}{2}\right)$	**M1**	Completes the square and integrates their constant term to the form $k \arctan(px + q)$. Condone slips in k, p or q.
	$\frac{7}{2} \arctan\left(\frac{x-2}{2}\right) + 2\ln(x^2 - 4x + 8) + c$	**A1**	Correct expression in the given form, or $A = \frac{7}{2}$ stated.
	Total for Question 10	**4**	

Question	Scheme	Marks	Notes
11(a)	$x^3-3x^2+9x+13=(x+1)(x^2-4x+13)$ $2x^2-3x+31\equiv P(x^2-4x+13)+Q(x+1)$	**M1**	Uses the factorised denominator and multiplies through to form an identity in P and Q. Condone a sign slip.
	$x^2\colon P=\dots$ $x\colon -4P+Q=-3\Rightarrow Q=\dots$	**dM1**	Compares coefficients or substitutes values of x, and solves two equations for both P and Q.
	$P=2,\ Q=5$	**A1**	
		[3]	
11(b)	$\int\frac{\text{“2”}}{x+1}\,\mathrm{d}x=2\ln\lvert x+1\rvert$	**M1 A1ft**	**M1** Integrates their $\frac{P}{x+1}$ to $k\ln\lvert x+1\rvert$. **A1ft** $2\ln\lvert x+1\rvert$, ft their P.
	$x^2-4x+13=(x-2)^2+9$ $\int\frac{\text{“5”}}{(x-2)^2+9}\,\mathrm{d}x=k\arctan\left(\frac{x-2}{3}\right)$	**M1**	Completes the square as $(x-2)^2+a$, $a>0$, and integrates their term to $k\arctan\left(\frac{x-2}{\sqrt{a}}\right)$. Condone slips in k.
	$\int g(x)\,\mathrm{d}x=2\ln\lvert x+1\rvert$ $+\frac{5}{3}\arctan\left(\frac{x-2}{3}\right)+c$	**A1ft**	$B=\frac{5}{3}$, ft their Q as $\frac{Q}{3}$. Condone a missing $+c$.
		[4]	
11(c)	$\int_0^\infty g(x)\,\mathrm{d}x=\lim_{t\to\infty}\int_0^t g(x)\,\mathrm{d}x$ $\int_0^t g(x)\,\mathrm{d}x=2\ln(t+1)+\frac{5}{3}\arctan\left(\frac{t-2}{3}\right)$ $-\frac{5}{3}\arctan\left(-\frac{2}{3}\right)$	**M1**	Replaces the upper limit by a variable, or writes a limit, and substitutes it and 0 into their (b) answer of the form $A\ln\lvert x+1\rvert+B\arctan(\dots)$.
	$t\to\infty\colon\ \arctan\left(\frac{t-2}{3}\right)\to\frac{\pi}{2},\ 2\ln(t+1)\to\infty$ so the integral diverges	**A1 AG**	cso. States the arctan term tends to $\frac{\pi}{2}$ or is bounded, $\ln(t+1)\to\infty$, and concludes divergence. A0 for evaluating at ∞ directly.
		[2]	
	Total for Question 11	**9**	

Question	**Scheme**	**Marks**	**Notes**
12(i)	$\cosh y = x \Rightarrow \sinh y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$	**M1**	Writes $\cosh y = x$ and differentiates implicitly, or finds $\frac{\mathrm{d}x}{\mathrm{d}y}$ and inverts. Condone a sign slip.
	$\sinh^2 y = \cosh^2 y - 1 = x^2 - 1$	**M1**	Uses $\cosh^2 y - \sinh^2 y = 1$ to express $\sinh y$ or $\sinh^2 y$ in terms of x. Condone a sign slip in the identity.
	$y \geq 0 \Rightarrow \sinh y = \sqrt{x^2-1}$ $\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}}$	**A1 AG**	Printed result without error, with the positive root justified, e.g. $y \geq 0$ so $\sinh y \geq 0$.
		[3]	
12(ii)	$y = \int \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x$ $= \int \frac{x+\sqrt{x^2-1}}{\sqrt{x^2-1}\left(x+\sqrt{x^2-1}\right)}\,\mathrm{d}x$	**M1**	Multiplies numerator and denominator of the integrand by $x + \sqrt{x^2-1}$.
	$= \int \frac{1+\frac{x}{\sqrt{x^2-1}}}{x+\sqrt{x^2-1}}\,\mathrm{d}x$, numerator the derivative of the denominator	**A1**	Integrand in the form $\frac{f'(x)}{f(x)}$ with $f(x) = x + \sqrt{x^2-1}$, the derivative shown or stated.
	$y = \ln\left(x + \sqrt{x^2-1}\right) + c$ $x = 1,\ y = 0 \Rightarrow 0 = \ln 1 + c \Rightarrow c = \ldots$	**dM1**	Integrates to $\ln\left(x + \sqrt{x^2-1}\right) + c$ and substitutes $x = 1$, $y = \operatorname{arcosh} 1 = 0$ to find c.
	$c = 0 \Rightarrow y = \ln\left(x + \sqrt{x^2-1}\right)$	**A1 AG**	Printed result without error from a correct evaluation of c. A0 if c is omitted or assumed zero.
		[4]	
12(iii)	$\sqrt{4x^2-9} = 3\sqrt{\left(\frac{2x}{3}\right)^2 - 1}$ $\Rightarrow \int \frac{1}{\sqrt{4x^2-9}}\,\mathrm{d}x = k \operatorname{arcosh}\left(\frac{2x}{3}\right)$	**M1**	Rewrites the integrand to use (i), by factorising or a substitution such as $u = \frac{2x}{3}$, and integrates to $k \operatorname{arcosh}(px)$. Condone wrong constants.
	$\frac{1}{2}\operatorname{arcosh}\left(\frac{2x}{3}\right)$	**A1**	
	$\frac{1}{2}\operatorname{arcosh}\left(\frac{2}{3} \times \frac{5}{2}\right) - \frac{1}{2}\operatorname{arcosh}\left(\frac{2}{3} \times \frac{15}{8}\right)$	**dM1**	Substitutes both limits into their $k \operatorname{arcosh}(px)$ and subtracts, in the correct order.
	$= \frac{1}{2}\left(\operatorname{arcosh}\frac{5}{3} - \operatorname{arcosh}\frac{5}{4}\right)$	**A1**	Printed result, with no errors seen.
	$\operatorname{arcosh}\frac{5}{3} = \ln\left(\frac{5}{3} + \sqrt{\frac{25}{9} - 1}\right)\ \ [= \ln 3]$ $\operatorname{arcosh}\frac{5}{4} = \ln\left(\frac{5}{4} + \sqrt{\frac{25}{16} - 1}\right)\ \ [= \ln 2]$	**M1**	Uses (ii) to write both $\operatorname{arcosh}\frac{5}{3}$ and $\operatorname{arcosh}\frac{5}{4}$ as logarithms. Need not be simplified.

Question 12 continued

Question	**Scheme**	**Marks**	**Notes**
	$\frac{1}{2}(\ln 3 - \ln 2) = \frac{1}{2}\ln\frac{3}{2}$	**A1**	Any exact logarithmic form, e.g. $\frac{1}{2}(\ln 3 - \ln 2)$ or $\ln\sqrt{\frac{3}{2}}$. A decimal alone scores A0.
		[6]	
	Total for Question 12	**13**	

Question	Scheme	Marks	Notes
13(a)	$25x^2 - 20x + 13 = (5x-2)^2 + 9$	**M1**	Completes the square to $(5x-2)^2 \pm k$ or $25\left(x - \frac{2}{5}\right)^2 \pm k$, $k \neq 0$. Condone an arithmetic slip in k.
	$\int \frac{\mathrm{d}x}{(5x-2)^2+9} = \frac{1}{15}\arctan\left(\frac{5x-2}{3}\right)$	**M1 A1**	**M1** Integrates their completed square to $\lambda \arctan\left(\frac{5x-2}{\mu}\right)$, or the same in $x - \frac{2}{5}$ or a substituted variable. Condone a wrong λ.
	$\frac{1}{15}\left(\arctan 1 - \arctan(-1)\right)$	**dM1**	Substitutes both limits into their arctan form and subtracts, or the changed limits after a substitution.
	$= \frac{\pi}{30}$	**A1**	cso.
		[5]	
13(b)	$\int \frac{\mathrm{d}x}{\sqrt{(5x-2)^2+9}}$ $= \lambda \ln\left(5x - 2 + \sqrt{(5x-2)^2+9}\right)$	**M1 A1**	**M1** Integrates to $\lambda \operatorname{arsinh}\left(\frac{u}{\mu}\right)$ or $\lambda \ln\left(u + \sqrt{u^2+\mu^2}\right)$, with u as $5x-2$, $x - \frac{2}{5}$ or a substituted variable. Condone a wrong λ. **A1** Correct inside, $\frac{u}{3}$ or $u + \sqrt{u^2+9}$ with $u = 5x-2$, or the equivalent in their variable.
	$\lambda = \frac{1}{5}$	**A1**	
	$\frac{1}{5}\left(\ln(3+3\sqrt{2}) - \ln(-3+3\sqrt{2})\right)$	**dM1 A1**	**dM1** Substitutes both limits into their integral and subtracts, or the changed limits after a substitution. **A1** oe, such as $\frac{1}{5}\left(\operatorname{arsinh} 1 - \operatorname{arsinh}(-1)\right)$.
	$\frac{1}{5}\ln\left(\frac{1+\sqrt{2}}{\sqrt{2}-1}\right)$ $= \frac{1}{5}\ln\left(\frac{(1+\sqrt{2})(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)}\right)$	**M1**	Combines their logarithms and rationalises, or uses $\ln(\sqrt{2}-1) = -\ln(\sqrt{2}+1)$, or writes arsinh 1 as $\ln(1+\sqrt{2})$. Condone sign slips.
	$= \frac{1}{5}\ln(3+2\sqrt{2})$	**A1**	Or $\frac{2}{5}\ln(1+\sqrt{2})$, or any correct rational p, q, r.
		[7]	
	Total for Question 13	**12**	

Question	Scheme	Marks	Notes
14(i)	$\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}} = \frac{e^{2y}-1}{e^{2y}+1}$	**M1 A1**	**M1** Writes $\tanh y$ in exponentials and multiplies numerator and denominator by e^y, or equivalent. Condone sign slips. **A1** oe, such as $\frac{1-e^{-2y}}{1+e^{-2y}}$.
	$x(e^{2y}+1) = e^{2y} - 1 \Rightarrow e^{2y} = \dots$ $\Rightarrow 2y = \ln(\dots)$	**M1**	Sets their expression equal to x, makes e^{2y} (or e^{-2y}) the subject and takes logarithms. Condone sign slips.
	$e^{2y}(1-x) = 1 + x \Rightarrow e^{2y} = \frac{1+x}{1-x}$ $y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$ $\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$	**A1 AG**	Correct equation for e^{2y} leading without error to the printed result, and the artanh form stated. A0 if the artanh line is missing.
		[4]	
14(ii)	$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{artanh} x) = \frac{1}{2}\left(\frac{1}{1+x} + \frac{1}{1-x}\right)$	**M1**	Differentiates the logarithmic form from (i) to two terms $\frac{a}{1 \pm x}$, or uses partial fractions, or the standard $\int \frac{1}{a^2 - x^2}\,\mathrm{d}x$ with $a = \frac{2}{3}$. Condone sign slips.
	$\int \frac{1}{1-x^2}\,\mathrm{d}x = \operatorname{artanh} x \ (+c)$	**A1**	oe, such as $\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$, or the correct partial fractions, or the correct a and factor.
	$\int \frac{1}{4-9x^2}\,\mathrm{d}x = \frac{1}{4}\int \frac{1}{1-\left(\frac{3x}{2}\right)^2}\,\mathrm{d}x$ $= k \operatorname{artanh}\left(\frac{3x}{2}\right)$	**M1**	Takes out the factor 4, or substitutes $u = \frac{3x}{2}$, and integrates to $k \operatorname{artanh}\left(\frac{3x}{2}\right)$ or $k\ln\left(\frac{2+3x}{2-3x}\right)$, any k.
	$\frac{1}{6}\operatorname{artanh}\left(\frac{3x}{2}\right)$	**A1**	oe, such as $\frac{1}{12}\ln\left(\frac{2+3x}{2-3x}\right)$.
	$\frac{1}{12}\left[\ln\left(\frac{1+\frac{3x}{2}}{1-\frac{3x}{2}}\right)\right]_{1/5}^{2/5}$ $= \frac{1}{12}\left(\ln 4 - \ln\frac{13}{7}\right)$	**dM1**	Substitutes both limits into a logarithmic form of their integral and subtracts.
	$\frac{1}{12}\ln\left(\frac{28}{13}\right)$	**A1**	Any exact logarithmic form, such as $\frac{1}{12}(\ln 28 - \ln 13)$. A decimal, awrt 0.0639, scores A0.
		[6]	
	Total for Question 14	**10**	

Question	**Scheme**	**Marks**	**Notes**
15	$\frac{P}{x-1}+\frac{Qx+R}{x^2+4x+13}$ $x^2+14x+21 \equiv P(x^2+4x+13) + (Qx+R)(x-1)$	**M1**	Uses the correct partial-fraction form and multiplies through to an identity. Condone slips in expanding.
	$P+Q=1,\ 4P-Q+R=14,\ 13P-R=21$ $\Rightarrow P=\dots,\ Q=\dots,\ R=\dots$	**dM1**	Compares coefficients or substitutes values of x, and solves for all three constants. Condone arithmetic slips.
	$P=2,\ Q=-1,\ R=5$	**A1**	
	$-x+5=-\frac{1}{2}(2x+4)+7$ $\int \frac{2}{x-1}\,\mathrm{d}x - \frac{1}{2}\int \frac{2x+4}{x^2+4x+13}\,\mathrm{d}x$ $= \dots \ln\lvert x-1\rvert + \dots \ln(x^2+4x+13)$	**M1**	Writes their $Qx+R$, $Q\neq 0$, as $k(2x+4)+m$ and integrates to $\alpha\ln\lvert x-1\rvert+\beta\ln(x^2+4x+13)$. Condone coefficient slips.
	$2\ln\lvert x-1\rvert - \frac{1}{2}\ln(x^2+4x+13)$	**A1**	Correct $A=2$ and $B=-\frac{1}{2}$.
	$x^2+4x+13=(x+2)^2+9$ $7\int \frac{1}{(x+2)^2+9}\,\mathrm{d}x = \dots \arctan\left(\frac{x+2}{3}\right)$	**M1**	Completes the square and integrates their constant m over $(x+2)^2+9$ to $k\arctan\left(\frac{x+2}{3}\right)$. Condone slips in k.
	$\dots + \frac{7}{3}\arctan\left(\frac{x+2}{3}\right)+c$ $A=2,\ B=-\frac{1}{2},\ D=\frac{7}{3}$	**A1**	Correct $D=\frac{7}{3}$, or awrt 2.33.
	Total for Question 15	**7**	

Question	**Scheme**	**Marks**	**Notes**
16(i)	$1-\tanh^2 u = 1-\left(\frac{e^u-e^{-u}}{e^u+e^{-u}}\right)^2$	**M1**	Writes $\tanh u$ and $\operatorname{sech} u$, or $\sinh u$ and $\cosh u$, in exponentials and attempts the identity from either side. Condone sign slips.
	$=\frac{(e^u+e^{-u})^2-(e^u-e^{-u})^2}{(e^u+e^{-u})^2}$ $=\frac{4}{(e^u+e^{-u})^2}$	**A1**	Numerator simplified to 4, oe such as $\cosh^2 u-\sinh^2 u=1$ shown in exponentials.
	$=\left(\frac{2}{e^u+e^{-u}}\right)^2=\operatorname{sech}^2 u$	**A1 AG**	Correct exponential working leading without error to $\operatorname{sech}^2 u$. Hyperbolic identities quoted without exponentials score A0.
		[3]	
16(ii)	$y=\tanh u=\frac{e^u-e^{-u}}{e^u+e^{-u}} \Rightarrow y=\frac{e^{2u}-1}{e^{2u}+1}$	**M1 A1**	**M1** Writes $y=\tanh u$ in exponentials and clears e^{-u}, such as by multiplying through by e^u. Condone sign slips. **A1** oe such as $ye^{2u}+y=e^{2u}-1$.
	$e^{2u}(1-y)=1+y \Rightarrow e^{2u}=\frac{1+y}{1-y}$ $\Rightarrow 2u=\ln\left(\frac{1+y}{1-y}\right)$	**M1**	Collects the e^{2u} terms, makes e^{2u} the subject and takes logarithms. Condone sign slips.
	$\operatorname{artanh} y=\frac{1}{2}\ln\left(\frac{1+y}{1-y}\right)$	**A1 AG**	Correct working leading without error to the printed result, with u replaced by $\operatorname{artanh} y$. No reference to $\lvert y\rvert<1$ needed.
		[4]	
16(iii)	$x=4\tanh u \Rightarrow \mathrm{d}x=4\operatorname{sech}^2 u\,\mathrm{d}u$ $\int\frac{1}{16-x^2}\,\mathrm{d}x=\int\frac{4\operatorname{sech}^2 u}{16-16\tanh^2 u}\,\mathrm{d}u$	**M1**	Substitutes $x=4\tanh u$ and $\mathrm{d}x=k\operatorname{sech}^2 u\,\mathrm{d}u$ to obtain an integral wholly in u.
	$=\int\frac{4\operatorname{sech}^2 u}{16\operatorname{sech}^2 u}\,\mathrm{d}u=\int\frac{1}{4}\,\mathrm{d}u$	**A1**	
	$=\frac{u}{4}\ (+c),\ \tanh u=\frac{x}{4}\Rightarrow u=\dots$	**M1**	Integrates their constant integrand to ku and uses $\tanh u=\frac{x}{4}$ to return to x.
	$\int\frac{1}{16-x^2}\,\mathrm{d}x=\frac{1}{4}\operatorname{artanh}\frac{x}{4}+c$	**A1**	From correct working, with $+c$.
	$\operatorname{artanh}\frac{x}{4}=\frac{1}{2}\ln\left(\frac{1+\frac{x}{4}}{1-\frac{x}{4}}\right)$	**M1**	Uses (ii) with $y=\frac{x}{4}$ in their $k\operatorname{artanh}\frac{x}{4}$.

Question 16 continued

Question	Scheme	Marks	Notes
	$\int \frac{1}{16-x^2}\,\mathrm{d}x = \frac{1}{4} \times \frac{1}{2} \ln\left(\frac{4+x}{4-x}\right) + c$ $= \frac{1}{8} \ln\left(\frac{4+x}{4-x}\right) + c$	**A1 AG**	Clears the fractions inside the logarithm and combines the coefficients without error to the printed result.
		[6]	
16(iv)	$-t^2 + 6t + 7 = -(t-3)^2 + \dots$	**M1**	Reaches a form $a - (t \pm 3)^2$ or $-\left((t \pm 3)^2 + b\right)$. Condone slips in the constant.
	$-t^2 + 6t + 7 = 16 - (t-3)^2$	**A1**	
	$\int \frac{1}{16-(t-3)^2}\,\mathrm{d}t = \frac{1}{8} \ln\left(\frac{4+(t-3)}{4-(t-3)}\right)$	**M1**	Integrates their $\frac{1}{a-(t-3)^2}$ as in (iii), with $x = t-3$, to $k \ln\left(\frac{\sqrt{a}+t-3}{\sqrt{a}-t+3}\right)$ or $k\,\mathrm{artanh}\,\frac{t-3}{\sqrt{a}}$.
	$\left[\frac{1}{8} \ln\left(\frac{t+1}{7-t}\right)\right]_1^5$ $= \frac{1}{8} \ln 3 - \frac{1}{8} \ln \frac{1}{3}$	**dM1**	Substitutes the limits 5 and 1 into their integral and subtracts, or 2 and -2 after changing variable to $t-3$.
	$= \frac{1}{4} \ln 3$	**A1 AG**	Exact working leading without error to $\frac{1}{4}\ln 3$. A decimal check alone scores A0.
		[5]	
Total for Question 16		**18**	

Question	Scheme	Marks	Notes
17(i)	$x^2-6x+5=(x-3)^2-4$ $\int \frac{1}{\sqrt{(x-3)^2-4}}\,\mathrm{d}x = \operatorname{arcosh}\left(\frac{x-3}{\cdots}\right)$	**M1**	Completes the square to $(x-3)^2-k$ and integrates to the form arcosh $\left(\frac{x-3}{p}\right)$, by standard result or substitution. Condone a wrong k.
	$\operatorname{arcosh}\left(\frac{x-3}{2}\right)+c$	**A1**	
	$\ln\left(\frac{x-3}{2}+\sqrt{\frac{(x-3)^2}{4}-1}\right)+c$	**A1ft**	ft their arcosh $\left(\frac{x-3}{p}\right)$, using $\operatorname{arcosh} t=\ln\left(t+\sqrt{t^2-1}\right)$. oe, including $\ln\left(x-3+\sqrt{x^2-6x+5}\right)+c$.
		[3]	
17(ii)	$x-3=2\cosh u \Rightarrow \mathrm{d}x=2\sinh u\,\mathrm{d}u$ $\sqrt{x^2-6x+5}=\sqrt{4\cosh^2 u-4}$ $=2\sinh u$	**M1**	Replaces $\mathrm{d}x$ by $k\sinh u\,\mathrm{d}u$ and the square root by an expression in u. Condone coefficient slips.
	$\int\sqrt{x^2-6x+5}\,\mathrm{d}x=4\int\sinh^2 u\,\mathrm{d}u$	**A1**	oe, such as $\int(4\cosh^2 u-4)\,\mathrm{d}u$.
	$2\int(\cosh 2u-1)\,\mathrm{d}u=\ldots$	**M1**	Uses $\cosh 2u=1\pm2\sinh^2 u$, or exponentials, and integrates to the form $p\sinh 2u+qu$ or its exponential equivalent. Condone sign and coefficient slips.
	$\sinh 2u-2u+c$	**A1**	oe
	$\sinh 2u=2\cosh u\sqrt{\cosh^2 u-1}$ $=(x-3)\sqrt{\frac{(x-3)^2}{4}-1}$ $u=\ln\left(\frac{x-3}{2}+\sqrt{\frac{(x-3)^2}{4}-1}\right)$	**M1**	Uses $\sinh 2u=2\sinh u\cosh u$ with $\sinh u=\sqrt{\cosh^2 u-1}$, and replaces u by their logarithm from (i), all in terms of x.
	$I=(x-3)\sqrt{\frac{(x-3)^2}{4}-1}-2u+c$ $=2\left(\frac{x-3}{2}\sqrt{\frac{(x-3)^2}{4}-1}\right.$ $\left.-\ln\left(\frac{x-3}{2}+\sqrt{\frac{(x-3)^2}{4}-1}\right)\right)+c$	**A1 AG**	cso. Correct result in u converted to the printed result without error.
		[6]	
	Total for Question 17	**9**	